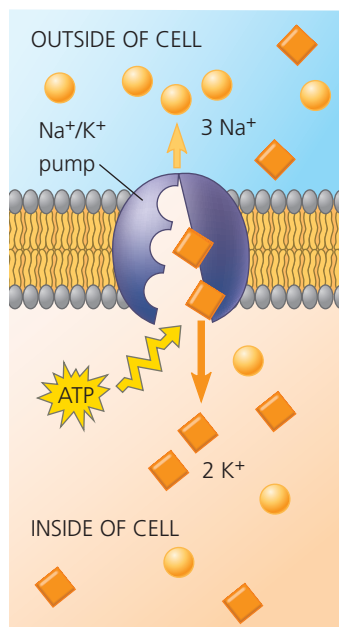


Formation of the Resting Potential

Potassium ions (K^+) and sodium ions (Na^+) play an essential role in the formation of the resting potential. These ions each have a concentration gradient across the plasma membrane of a neuron (**Table 48.1**). In most neurons, the concentration of K^+ is higher inside the cell, while the concentration of Na^+ is higher outside. The Na^+ and K^+ concentration gradients are maintained by the **sodium-potassium pump**. This pump uses the energy of ATP hydrolysis to actively transport Na^+ out of the cell and K^+ into the cell. (There are also concentration gradients for chloride ions (Cl^-) and other anions, as shown in Table 48.1, but we can ignore these for now.)

Ion	Intracellular Concentration (mM)	Extracellular Concentration (mM)
Potassium (K^+)	140	5
Sodium (Na^+)	15	150
Chloride (Cl^-)	10	120
Large anions (A^-), such as proteins, inside cell	100	Not applicable

The sodium-potassium pump transports three Na^+ out of the cell for every two K^+ that it transports in (**Figure 48.6**). Although this pumping generates a net export of positive charge, the pump acts slowly. The resulting change in the membrane potential is therefore quite small—only a few millivolts. Why, then, is there a membrane potential of -60 to -80 mV in a resting neuron? The answer lies in ion movement through **ion channels**, pores formed by clusters of specialized proteins that span the membrane. Ion channels allow ions to diffuse back and forth across the membrane. As ions diffuse through channels, they carry with them units of electrical



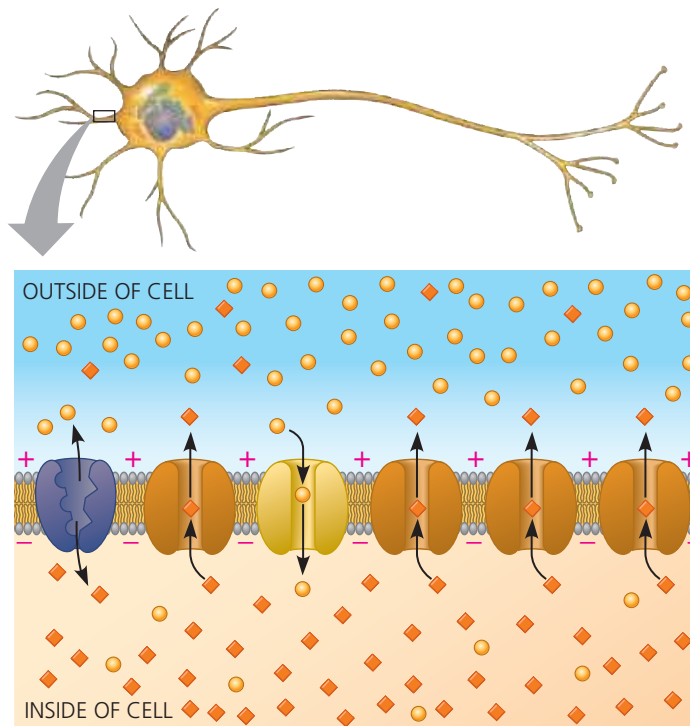
◀ **Figure 48.6** Summary of active transport by the sodium-potassium pump. You can find a step-by-step description of the pump's activity in Figure 7.15.

charge. Furthermore, ions can move quite rapidly through ion channels. When this occurs, the resulting current—a *net* movement of positive or negative charge—generates a membrane potential, or voltage across the membrane.

Concentration gradients of ions across a plasma membrane represent a chemical form of potential energy that can be harnessed for cellular processes (see Figure 44.17). In neurons, the ion channels that convert this chemical potential energy to electrical potential energy can do so because they have *selective permeability*, allowing only certain ions to pass. For example, a potassium channel allows K^+ to diffuse freely across the membrane, but not other ions, such as Na^+ or Cl^- .

Diffusion of K^+ through potassium channels that are always open (sometimes called *leak channels*) is critical for establishing the resting potential (**Figure 48.7**). The K^+

▼ **Figure 48.7** The basis of the membrane potential. The sodium-potassium pump generates and maintains the concentration gradients of Na^+ and K^+ shown in Table 48.1. The $[Na^+]$ gradient results in very little net diffusion of Na^+ in a resting neuron because very few sodium channels are open. In contrast, the many open potassium channels allow a significant net outflow of K^+ . Because the membrane is only weakly permeable to chloride and other anions, this outflow of K^+ results in a net negative charge inside the cell.



Key



? The potassium and sodium channels have the same rough overall structure, as shown. How must these proteins differ to allow passage of only a particular ion?

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